Offset[Len]   Field Name and Description   Data Type

RECORD TTF

- Identification

  The Teller Transaction File (TTF) contains a record for each transaction code and the type of account affected by the transaction. It is a complete table of all the transactions and account types supported by the BASE24-teller product. TTF records determine whether a transaction is supported, the type of account or accounts that are affected by the transaction, whether the level of the teller terminal operator is sufficient to perform the transaction, and which process handles the authorization of the transaction. One record is required in the TTF for each transaction code processed by an institution.

- Security

  The TTF is secured under Tandem's group level security so that only authorized personnel can access the file. This file can only be accessed via an AFT terminal with proper operator signon. The specific security for the TTF should be restricted to a high level because changing the TTF will affect how the BASE24-teller System processes a specific transaction. The operator must have super group (255) security to modify TTF records in FIID 0000 or to add/delete TTF records by FIID.

```
User Identification - <Base24 node name>.*
File Code           - 0
Access Security     - GGGG
```

- Usage

  The TTF is read into extended memory by the BASE24-teller Device Handler and Authorization processes during startup and is used to determine how to process specific transaction codes. The internal table maintained by the Device Handler contains information related to whether the transaction is supported by the terminal owner and whether interbank routing is supported by the terminal owner. The internal table maintained by the Authorization process

```
Offset[Len]   Field Name and Description                      Data Type

                contains information related to whether the
                transaction is supported by the account owner
                for on-us or not-on-us transactions, the
                authorization level, whether completions
                are sent, etc.  The BASE24 AFT System
                accesses the actual file but any changes
                made are not reflected in processing
                until the next startup or warmboot of the
                Device Handler and Authorization processes.

                When ACI installs BASE24-teller, a full
                set of TTF records is placed in the TTF
                under an FIID of 0000.  This full set
                called the default TTF, must then be
                copied for each institution in the system,
                using that institution's FIID.

                Authorization  Input Sequential   Shared
                Device Handler Input Sequential   Shared
                File Maint     I/O   Random ( )   Shared
                Reports        Input Sequential   Shared

              - Organization

                The TTF is a key sequenced file with fixed
                length records.  The key to the file is FIID,
                transaction code, from account type and to
                account type.


0      [80]

0      [10]   prikey

              The values in the following fields form the primary key
              to records in the TTF.

0      [4]    prikey.fiid.byte[0:3]
                               [FIID]                          PIC X(4).

              The FIID of the financial institution defining the
              transaction.

4      [2]    prikey.tran^cde.byte[0:1]                        PIC X(2).

              The transaction code of the transaction being defined.
              This code, together with a from account type and to
              account type, represents a single type of transaction.
              Valid values are:

              10    = Withdrawal
              11    = Cash Check Transit
              12    = Cash Official Check
              13    = Cash Certificate of Deposit
```

```
Offset[Len]   Field Name and Description                    Data Type

              14    = Cash Bond
              15    = Cash Coupon
              16    = Miscellaneous Debit
              17    = Cash Affiliate/Correspondent Check
              18    = Debit Memo Post
              20    = Deposit
              21    = Split Deposit
              22    = Miscellaneous Credit
              23    = Credit Memo Post
              30    = PBF Inquiry
              31    = PBF Short Inquiry (Ledger and Available
                      Balance)
              32    = SPF Inquiry
              33    = CAF Inquiry
              34    = NBF Inquiry
              35    = Passbook Print
              36    = Passbook Reprint
              37    = WHFF Inquiry
              40    = Transfer
              50    = Payment
              60    = Purchase Money Order
              61    = Purchase Cashier's Check
              62    = Purchase Traveler's Check
              63    = Purchase Bank Draft
              64    = Purchase Certified Check
              65    = Purchase Bond
              66    = Purchase Miscellaneous
              73    = Change Card Status on CAF
              74    = Change Account Status on CAF/PBF
              75    = PIN Verification
              80    = Add Stop Payment to SPF
              81    = Delete Stop Payment from SPF
              82    = Change Account Status on PBF
              83    = Change PBF Stop Pay/Warning Status
              84    = Add Warning to WHFF
              85    = Add Hold to WHFF
              86    = Delete Hold from WHFF
              87    = Add Deposit Float to WHFF
              88    = Delete Deposit Float from WHFF
              89    = Delete Warning from WHFF
              90    = Logon - Start of Business Day
              91    = Logoff - End of Business Day
              92    = Sign On - Start of Balancing Day
              93    = Sign Off - End of Balancing Day
              94-99 = Reserved for Administrative Transactions

6     [2]     prikey.from^acct^typ.byte[0:1]                PIC X(2).

              The from account type of the transaction being defined.
              This typically indicates the type of account that is
              debited by the transaction.  Valid values are:

              00 = Not Applicable
              01 = DDA (Checking Account)
```

```
Offset[Len]   Field Name and Description                     Data Type

              11 = Savings
              12 = IRA
              13 = Certificate of Deposit
              21 = NOW
              31 = Credit Card
              32 = Credit Line
              41 = Installment Loan
              42 = Mortgage Loan
              43 = Commercial Loan
              50 = Utility Payment
              51 = Utility1 Payment
              52 = Utility2 Payment
              53 = Utility3 Payment
              54 = Utility4 Payment
              55 = Utility5 Payment


8     [2]     prikey.to^acct^typ.byte[0:1]                   PIC X(2).

              The to account type of the transaction being defined.
              This typically indicates the type of account that is
              credited by the transaction.  Valid values are:

              00 = Not Applicable
              01 = DDA (Checking Account)
              11 = Savings
              12 = IRA
              13 = Certificate of Deposit
              21 = NOW
              31 = Credit Card
              32 = Credit Line
              41 = Installment Loan
              42 = Mortgage Loan
              43 = Commercial Loan
              50 = Utility Payment
              51 = Utility1 Payment
              52 = Utility2 Payment
              53 = Utility3 Payment
              54 = Utility4 Payment
              55 = Utility5 Payment

10    [1]     auth^lvl                                       PIC X(1).

              A code indicating the authorization level of the
              transaction from the the perspective of the account
              owner.  Valid values are:

              0 = Transaction Not Supported
              1 = Online Authorization, Host Authorizes
              2 = Offline Authorization, BASE24 Authorizes
              3 = Reserved for future use
              4 = Log Only, BASE24-teller Authorization Process
                  logs transaction to the TTLF
```

Offset[Len]   Field Name and Description   Data Type

The Positive Balance File (PBF) on BASE24 is only updated for transactions with an AUTH-LVL value of 2 (Offline Authorization).

11   [1]   min^tlr^lvl   PIC X(1).

A code indicating the minimum teller level that is required in order to perform this particular transaction. Any teller level equal to or greater than this minimum teller level is accepted as a valid teller level to perform this transaction. Valid values are:

0 = No teller level required. This transaction does not require a particular teller level.
1 = Teller level. The transaction can be performed if entered by a teller.
2 = Supervisor level. The transaction can be performed if entered by a supervisor.
3 = Manager level. The transaction can be performed if entered by a manager.
4 = Restricted. The transaction is denied and cannot be performed by any teller level.

12   [1]   whff^search^req   PIC X(1).

A code identifying whether BASE24 is required to search the WHFF for this transaction. This field may only be set to a value of Y for financial transactions. Valid values are:

Y = Yes, search the WHFF on this transaction.
N = No, do not search the WHFF on this transaction.

13   [1]   paperless^tran   PIC X(1).

A code indicating how the institution handles a transaction when the Teller Transaction Log File (TTLF) extract is run. The value in this field is used to assist the host software. Valid values are:

Y = Yes, this is a paperless transaction. The institution should post the transaction to the host from the extract file.
N = No, this is not a paperless transaction. The institution should use a paper source document to post the transaction to the host.

14   [6]   prnt^descr.byte[0:5]   PIC X(6).

A short version of the transaction description that is printed in the passbook to identify the transaction.

```
Offset[Len]   Field Name and Description                     Data Type

20    [1]     compl^req                                      PIC X(1).

              A code identifying the type(s) of transactions for
              which BASE24-teller generates a completion to
              the host.  Valid values are:

              N = No, do not send completion messages to the host.
              A = Yes, send completion messages to the host for
                  approved transactions only.
              D = Yes, send completion messages to the host for
                  denied transactions only.
              B = Yes, send completion messages to the host for
                  approved and denied transactions.

21    [30]    tran^descr.byte[0:29]                         PIC X(30).

              Displays a description of the transaction entered in
              the TRAN-CDE, FROM-ACCT-TYP, and TO-ACCT-TYP fields.
              The value in this field is for information only to
              ensure that the operator does not inadvertently change
              the wrong transaction.  Refer to the BASE24-teller
              Files Maintenance Manual for a list of the transaction
              descriptions contained in the TTF when ACI installs
              BASE24-teller.

51    [1]     proc^pre^auth                                  PIC X(1).

              Used by the Authorization process, this field indicates
              whether PBF pre-authorization holds should be taken into
              consideration when calculating account balances.  Valid
              values are:

              N = No, do not include pre-auth holds (default).
              Y = Yes, include pre-auth holds with balances.

52    [1]     term^tran^alwd                                 PIC X(1).

              Indicates whether this transaction is allowed
              by the terminal owner.  Valid values are:

              Y = Yes, this transaction is allowed by the
                  terminal owner.
              N = No, this transaction is not allowed by the
                  terminal owner.
              L = Log Only, this transaction is defined as
                  log only for the terminal owner.

53    [1]     not^on^us^tran^alwd                            PIC X(1).

              Indicates whether the account owner allows
              this transaction to be performed by their
              customer at a not-on-us terminal.  Valid values
              are:
```

```
Offset[Len]   Field Name and Description                        Data Type

              Y = Yes, this transaction is allowed by the
                  account owner at a not-on-us terminal
              N = No, this transaction is not allowed by the
                  account owner at a not-on-us terminal

54    [8]     user^fld1.byte[0:7]                               PIC X(8).

              Reserved for future use.

62    [18]    last^fm          [LAST-FM]

              This field describes the last on-line file maintenance
              update applied to this record.  This includes the user
              who did the update, the time at which it was done, the
              type of update and the terminal originating the update
              transaction.


62    [6]     last^fm.fm^timestamp[0:2]         TYPE BINARY 16 SIGNED.


              The date and time of the last file maintenence
              application.


68    [2]     last^fm.fm^user^grp               TYPE BINARY 16 SIGNED.

              The group number of the operator that performed the last
              file maintenance application.  This value is taken from
              the information given when the operator last logged on
              to BASE24.


70    [4]     last^fm.fm^user^num               TYPE BINARY 32 SIGNED.

              The user number of the operator that performed the last
              file maintenance application.  This value is taken from
              the information given when the operator last logged on
              to BASE24.


74    [1]     last^fm.updt^typ                                  PIC X(1).


              The type of file maintenance that was last applied to
              the record.  Valid values are:

              A =  Add
              B =  Change "before" image
              C =  Change "after" image
              D =  Delete
              S =  Logon security check
              O =  Entry into operator functions
```

Offset	[Len]	Field Name and Description	Data Type
75	[4]	last^fm.sta^num.byte[0:3] This field is not used.	PIC X(4).
79	[1]	last^fm.fill1	PIC X(1).
		END [size 80]	

```
Offset[Len]   Field Name and Description                  Data Type

RECORD TTLF


0     [516]

0     [108]   head              [HEAD]


              Identification

                  The Teller Transaction Log File (TTLF) contains a re
                  of every transaction that originated at a teller ter
                  during the business day.  The TTLF records are extra
                  daily to provide detailed transaction data to
                  participating institutions for processing at their h
                  computer systems.  All teller logon/logoff transacti
                  can be included, as well as all financial transactio
                  and all CAF, NBF, PBF, SPF and WHFF update and inqui
                  transactions from teller terminals.

              Security

                  The TTLF is secured under Tandem's group level secur
                  that only authorized personnel can access the file.
                  file is available in an on-line lookup mode through
                  terminal with a proper operator signon.

                  User Identification - <Base24 node name>.*
                  File Code           - 0
                  Access Security     - GGGG

              Usage

                  The Device Handler and Authorization processes log
                  transactions to the TTLF.  The TTLF is written
                  sequentially.  Each record written to the TTLF may
                  also contain one or more tokens.  The Token File
                  (TKN) controls which tokens are logged to the
                  TTLF for each record type (i.e. Financial,
                  File Inquiry/Update, Administrative).  For
                  File Inquiry/Update and Administrative record
                  types, the tokens logged can be controlled at
                  the subtype level.  For BASE24-teller, the subtype
                  is equivalent to the transaction code.  Records
                  in the TTLF can also be perused online via the
                  BASE24-teller AFT Subsystem.

                  Device Handler   Output Sequential  Shared
                  Authorization    Output Sequential  Shared
                  Settlement       Create
                  Reporting        Input Sequential
                  Extract          Input Sequential
                  Refresh          Input Sequential
                  File Look-up     Input Random       Shared
```

```
Offset[Len]   Field Name and Description                        Data Type

0      [8]      head.dat^tim                          TYPE BINARY 64 SIGNED.

                The JULIANTIMESTAMP of when the record was written
                to the TTLF.

8      [2]      head.rec^typ.byte[0:1]                           PIC X(2).

                A code indicating the general type of the record.
                Valid values are:

                01 = Financial Transaction
                02 = File Inquiry/Update Transaction
                03 = Reserved for Future Use
                04 = Administrative Transaction
                     Reflects the information from the logon, logoff,
                     sign on, and sign off transactions.

10     [4]      head.auth^ppd.byte[0:3]                          PIC X(4).

                The Authorization or Device Handler process that
                added the record to the TTLF by its PPD name.  The $
                sign is not included.  It is anticipated that this
                field will be used by various Performance Tools to
                be developed for BASE24-teller.

14     [30]     head.tlr

                The following fields contain the header data of each
                transaction written to the TTLF using keytag TR.

14     [4]      head.tlr.ln.byte[0:3]
                                 [LN]                            PIC X(4).

                The logical network of the financial institution whose
                terminal initiated the transaction.

18     [4]      head.tlr.fiid.byte[0:3]
                                 [FIID]                          PIC X(4).

                The FIID of the financial institution whose terminal
                initiated the transaction.

22     [8]      head.tlr.tlr^id.byte[0:7]                        PIC X(8).

                The teller identification information used to determine
                which teller unit initiated the transaction.

30     [6]      head.tlr.tran^dat
                                 [DAT]

                The date (YYMMDD) the transaction occurred.
```

```
Offset[Len]   Field Name and Description                       Data Type

30    [2]     head.tlr.tran^dat.yy.byte[0:1]                   PIC X(2).

32    [2]     head.tlr.tran^dat.mm.byte[0:1]                   PIC X(2).

34    [2]     head.tlr.tran^dat.dd.byte[0:1]                   PIC X(2).

36    [8]     head.tlr.tran^tim
                                [TIM]

              The time (HHMMSSTT) the transaction occurred.

36    [2]     head.tlr.tran^tim.hh.byte[0:1]                   PIC X(2).

38    [2]     head.tlr.tran^tim.mm.byte[0:1]                   PIC X(2).

40    [2]     head.tlr.tran^tim.ss.byte[0:1]                   PIC X(2).

42    [2]     head.tlr.tran^tim.tt.byte[0:1]                   PIC X(2).

14    [30]    head.tlr^r
                                [REDEFINES TLR]

14    [16]    head.tlr^r.fi^tlr

14    [4]     head.tlr^r.fi^tlr.ln.byte[0:3]
                                [LN]                           PIC X(4).

18    [4]     head.tlr^r.fi^tlr.fiid.byte[0:3]
                                [FIID]                         PIC X(4).

22    [8]     head.tlr^r.fi^tlr.tlr^id.byte[0:7]               PIC X(8).

30    [6]     head.tlr^r.tran^dat
                                [DAT]

30    [2]     head.tlr^r.tran^dat.yy.byte[0:1]                 PIC X(2).

32    [2]     head.tlr^r.tran^dat.mm.byte[0:1]                 PIC X(2).

34    [2]     head.tlr^r.tran^dat.dd.byte[0:1]                 PIC X(2).

36    [8]     head.tlr^r.tran^tim
                                [TIM]

36    [2]     head.tlr^r.tran^tim.hh.byte[0:1]                 PIC X(2).

38    [2]     head.tlr^r.tran^tim.mm.byte[0:1]                 PIC X(2).

40    [2]     head.tlr^r.tran^tim.ss.byte[0:1]                 PIC X(2).

42    [2]     head.tlr^r.tran^tim.tt.byte[0:1]                 PIC X(2).

44    [33]    head.acct
```

```
Offset[Len]   Field Name and Description                        Data Type


              The following fields contain the header data of each
              transaction written to the TTLF using keytag AC.

44    [4]     head.acct.ln.byte[0:3]
                               [LN]                             PIC X(4).

              The logical network of the account owner.

48    [4]     head.acct.fiid.byte[0:3]
                               [FIID]                           PIC X(4).

              The FIID of the account owner.

52    [19]    head.acct.acct^num.byte[0:18]                    PIC X(19).

              The application account number involved in the
              transaction.

71    [6]     head.acct.tran^typ^cde.byte[0:5]                  PIC X(6).

              The transaction code of the transaction.  This
              field contains the original transaction code
              which is not modified as a result of account
              type range support.

44    [33]    head.acct^r
                               [REDEFINES ACCT]



44    [27]    head.acct^r.fi^acct

44    [4]     head.acct^r.fi^acct.ln.byte[0:3]
                               [LN]                             PIC X(4).

48    [4]     head.acct^r.fi^acct.fiid.byte[0:3]
                               [FIID]                           PIC X(4).

52    [19]    head.acct^r.fi^acct.acct^num.byte[0:18]          PIC X(19).

71    [6]     head.acct^r.tran^typ^cde.byte[0:5]                PIC X(6).

77    [30]    head.crd

              The following fields contain the header data of each
              transaction written to the TTLF using keytag CR.

77    [4]     head.crd.ln.byte[0:3]
                               [LN]                             PIC X(4).

              The logical network of the card owner.

81    [4]     head.crd.fiid.byte[0:3]
```

```
Offset[Len]   Field Name and Description                       Data Type

                                 [FIID]                        PIC X(4).

              The FIID of the card owner.

85    [19]    head.crd.pan.byte[0:18]                         PIC X(19).

              The card number (PAN) involved in the transaction.

104   [3]     head.crd.mbr^num.byte[0:2]
                                 [MBR-NUM]                     PIC 9(3).

              The member number.  Right-justified and zero
              filled on the left.

77    [30]    head.crd^r
                                 [REDEFINES CRD]

77    [30]    head.crd^r.fi^crd

77    [4]     head.crd^r.fi^crd.ln.byte[0:3]
                                 [LN]                          PIC X(4).

81    [4]     head.crd^r.fi^crd.fiid.byte[0:3]
                                 [FIID]                        PIC X(4).

85    [19]    head.crd^r.fi^crd.crd^num.byte[0:18]            PIC X(19).

104   [3]     head.crd^r.fi^crd.mbr^num.byte[0:2]
                                 [MBR-NUM]                     PIC 9(3).

107   [1]     head.user^fld1                                   PIC X(1).

              Reserved for future use.

108   [80]    refr^hdr           [REFR-HDR]

108   [1]     refr^hdr.impact^ind                              PIC X(1).

              A code indicating whether the record should be
              considered when impacting a new refreshed set of
              account records.  Valid values are:

              0 = No, do not use the record for impacting.
              1 = Yes, use the record for impacting.


109   [1]     refr^hdr.user^fld3                               PIC X(1).

              Reserved for future use.

110   [2]     refr^hdr.amt^on^hld^ind[0:1]                     PIC X(1).

              These fields indicate how the AMT-ON-HLD field in
              the Base segment of the PBF should be impacted.  The
```

Offset[Len]   Field Name and Description   Data Type

amount on hold is impacted by the amount specified in REFR-HDR.AMT-ON-HLD. The first occurrence of this field applies to the "from" or first account involved in the transaction. The second occurrence of this field applies to the "to" or second account involved in the transaction. Valid values are:

```
I  =  Increment
D  =  Decrement
_  =  Not Impacted ( _ denotes a blank)
```

```
112   [16]      refr^hdr.amt^on^hld[0:1]          TYPE BINARY 64 SIGNED.
```

These fields contain the amount to be added to or subtracted from the AMT-ON-HLD field in the Base segment of the PBF. These fields are initialized for all types of transactions. The amount on hold is the total amount of account funds being held and not available to the customer. An example of held funds would be a deposit at the teller line consisting of checks. Although the funds might be credited to the current account balance, the customer could not make use of them until they were verified or cleared. The first occurrence of this field applies to the "from" or first account involved in the transaction. The second occurrence of this field applies to the "to" or second account involved in the transaction.

```
128   [2]       refr^hdr.avail^bal^ind[0:1]                    PIC X(1).
```

These fields indicate how the AVAIL-BAL field in the Base segment of the PBF should be impacted. These fields are initialized for all types of transactions. The available balance is impacted by the amount specified in REFR-HDR.AVAIL-BAL-AMT. The first occurrence of this field applies to the "from" or first account involved in the transaction. The second occurrence of this field applies to the "to" or second account involved in the transaction. Valid values are:

```
I  =  Increment
D  =  Decrement
_  =  Not Impacted ( _ denotes a blank)
```

```
130   [16]      refr^hdr.avail^bal^amt[0:1]       TYPE BINARY 64 SIGNED.
```

These fields contain the amount to be added to or subtracted from the AVAIL-BAL field in the Base segment of the PBF. These fields are initialized for all types of transactions. On approved "debit" type transactions, this amount is subtracted from the available balance. On approved "credit" type transactions, this amount is added to

```
Offset[Len]   Field Name and Description                      Data Type

              the available balance.  On approved "debit" reversal
              transactions, this amount is added to the
              available balance.  On approved "credit" reversal
              transactions, this amount is subtracted from
              the available balance.  The first occurrence of this
              field applies to the "from" or first account involved
              in the transaction.  The second occurrence of this field
              applies to the "to" or second account involved in the
              transaction.

146   [2]     refr^hdr.amt^dep^cr^ind[0:1]                    PIC X(1).

              These fields indicate how the AMT-DEP-CR field in the
              Teller segment of the PBF should be impacted.  These
              fields are initialized for all types of
              transactions.  The amount of deposit credit is
              impacted by the amount specified in REFR-HDR.AMT-DEP-CR.
              The first occurrence of this field applies to the
              "from" or first account involved in the transaction.
              The second occurrence of this field applies to the
              "to" or second account involved in the transaction.
              Valid values are:

              I  =  Increment
              D  =  Decrement
              _  =  Not Impacted ( _ denotes a blank)

148   [16]    refr^hdr.amt^dep^cr[0:1]          TYPE BINARY 64 SIGNED.

              These fields contain the amount to be added to or
              subtracted from the AMT-DEP-CR in the Teller
              segment of the PBF.  These fields are initialized
              for all transactions.  The first occurrence of this
              field applies to the "from" or first account
              involved in the transaction.  The second occurrence of
              this field applies to the "to" or second account
              involved in the transaction.

164   [2]     refr^hdr.ledg^cr^bal^ind[0:1]                   PIC X(1).

              These fields indicate how the LEDGER-BAL and CR-BAL
              fields in the Base segment of the PBF should be
              impacted.  The balances are impacted by the
              transaction amount(s) specified in the Financial
              Token.  The amount fields actually used will depend
              on the type of transaction.  The first occurrence of
              this field applies to the "from" or first account
              involved in the transaction.  The second occurrence of
              this field applies to the "to" or second account
              involved in the transaction.  Valid values are:

              I  =  Increment
              D  =  Decrement
              _  =  Not Impacted ( _ denotes a blank)
```

```
Offset[Len]   Field Name and Description                      Data Type


166   [2]     refr^hdr.cash^out^ind[0:1]                      PIC X(1).

              These fields indicate how the CASH-OUT-TODAY fields
              in the Base segment of the PBF should be impacted.
              Cash out today is impacted by the cash out
              amount specified in the Financial Token.  The
              first occurrence of this field applies to the "from"
              or first account involved in the transaction.  The
              second occurrence of this field applies to the "to"
              or second account involved in the transaction.
              Valid values are:

              I  =  Increment
              D  =  Decrement
              _  =  Not Impacted ( _ denotes a blank)

168   [2]     refr^hdr.cash^in^ind[0:1]                       PIC X(1).

              These fields indicate how the CASH-IN-TODAY fields
              in the Base segment of the PBF should be impacted.
              Cash in today is impacted by the cash in amount
              specified in the Financial Token.  The first
              occurrence of this field applies to the "from" or first
              account involved in the transaction.  The second
              occurrence of this field applies to the "to" or second
              account involved in the transaction.  Valid values are:

              I  =  Increment
              D  =  Decrement
              _  =  Not Impacted ( _ denotes a blank)

170   [8]     refr^hdr.tran^amt                  TYPE BINARY 64 SIGNED.

              The amount of the transaction.  This field has been
              defined in the refresh header so it would not be
              necessary to parse for the Financial Token in order
              to display the transaction amount on the TTLF Perusal
              Summary Screen.  This field is also used by BASE24-
              teller Report Programs.

178   [1]     refr^hdr.spf^refr^ind                           PIC X(1).

              A code indicating that the SPF has been impacted for
              Refresh.

179   [1]     refr^hdr.nbf^refr^ind                           PIC X(1).

              A code indicating that the NBF has been impacted for
              Refresh.

180   [1]     refr^hdr.whff^refr^ind                          PIC X(1).

              A code indicating that the WHFF has been impacted
```

```
Offset[Len]   Field Name and Description                     Data Type

              for Refresh.

181   [1]     refr^hdr.dda^refr^ind                          PIC X(1).

              A code indicating that the PBF for the DDA account has
              been impacted for Refresh.

182   [1]     refr^hdr.sav^refr^ind                          PIC X(1).

              A code indicating that the PBF for the savings account
              has been impacted for Refresh.

183   [1]     refr^hdr.ccd^refr^ind                          PIC X(1).

              A code indicating that the PBF for the credit card
              account has been impacted for Refresh.

184   [2]     refr^hdr.from^acct^passbook^ind[0:1]           PIC X(1).

              A flag indicating whether a record was added to, or
              deleted from the NBF.  This field is only used if
              the from/first account is a passbook account.  The first
              occurrence of this field is associated with the
              Financial Transaction being performed (e.g., Withdrawal
              from Savings, Deposit to Savings, etc.).  The second
              occurrence is used if the from account is a passbook
              account and a credit line/backup account was accessed to
              authorize the transaction.  For example, if a credit
              line/backup account is accessed to authorize a
              Withdrawal from Savings Transaction, the second
              occurrence of this field indicates whether a record
              was added to the NBF to reflect the transfer of funds
              from the credit line/backup account to the from account.
              Valid values are:

              I  =  The from account was a passbook account, and
                    the NBF record count was Incremented
              D  =  The from account was a passbook account, and
                    the NBF record count was Decremented
              _  =  The from account was not a passbook account, and
                    the NBF record count was not changed
                    ( _ denotes a blank)

186   [1]     refr^hdr.to^acct^passbook^ind                  PIC X(1).

              A flag indicating whether a record was added to, or
              deleted from the NBF.  This field is only used if the
              to/second account is a passbook account.  Valid values
              are:

              I  =  The to account was a passbook account, and
                    the NBF record count was Incremented
              D  =  The to account was a passbook account, and
                    the NBF record count was Decremented
```

```
Offset[Len]   Field Name and Description                      Data Type

              _  =  The to account was not a passbook account, and
                    the NBF record count was not changed
                    ( _ denotes a blank)

187   [1]     refr^hdr.cr^line^acct^passbook^ind              PIC X(1).

              A flag indicating whether a record was added to, or
              deleted from the NBF.  This field is only used if the
              credit line/backup account is a passbook account.
              Valid values are:

              I  =  The credit line/backup account was a passbook
                    account, and the NBF record count was Incremented
              D  =  The credit line/backup account was a passbook
                    account, and the NBF record count was Decremented
              _  =  The credit line/backup account was not a passbook
                    account, and the NBF record count was not changed
                    ( _ denotes a blank)

188   [328]   tstmh^data       [TSTMH-DATA]

              The TSTMH portion of the TTLF record.  This
              portion of the record contains transaction
              message data.


188   [42]    tstmh^data.sys

188   [4]     tstmh^data.sys.msg^typ.byte[0:3]                PIC 9(4).

              The type of message.  The following message types
              are supported:

              0210 = Financial Transaction Response
              0220 = Financial Transaction Advice
              0230 = Financial Transaction Advice Response
              0420 = Financial Transaction Reversal Advice
              0430 = Financial Transaction Reversal Response
              0310 = File Inquiry/Update Response
              0320 = File Update Advice
              0330 = File Update Advice Response
              0610 = Administrative Response
              0620 = Administrative Advice
              0630 = Administrative Advice Response

192   [1]     tstmh^data.sys.originator                       PIC X(1).

              A code indicating where the transaction request,
              reversal or advice originated.  Applicable
              to 0200, 0300, 0600, 0220, 0320, 0420 and 0620
              messages.  Valid values are:

              1 = Device controlled by BASE24
              2 = Device Handler
```

```
Offset[Len]   Field Name and Description                  Data Type

              3 = Authorization Process
              4 = Host Interface Process
              5 = Host

193   [1]     tstmh^data.sys.responder                    PIC X(1).

              A code indicating where the transaction response
              was generated.  Applicable to 0210, 0310,
              0610, 0230, 0330, 0430 and 0630 messages.  Valid
              values are:

              1 = Device controlled by BASE24
              2 = Device Handler
              3 = Authorization Process
              4 = Host Interface Process
              5 = Host

194   [18]    tstmh^data.sys.orig

              The following fields contain the transaction
              origination data.

194   [16]    tstmh^data.sys.orig.sta^name.byte[0:15]
                               [SYM-NAME]                PIC X(16).

              The name of the station sending the message.
              This field represents the terminal id.

210   [2]     tstmh^data.sys.orig.rte^stat.byte[0:1]      PIC 9(2).

              Indicates the status of a message at the system
              level.  If the value is 00, the response code
              field indicates the disposition of the message.
              This field is set by the Host Interface process
              and used by the Authorization process.  This
              field is reserved for the future development of
              online/offline processing.

212   [6]     tstmh^data.sys.tran^seq^num.byte[0:5]       PIC X(6).

              The sequence number of the transaction, typically
              generated by the BASE24-teller Device Handler.

218   [6]     tstmh^data.sys.dev^seq^num.byte[0:5]        PIC 9(6).

              The sequence number supplied by the device and printed
              on the receipt.

224   [6]     tstmh^data.sys.trace^num.byte[0:5]          PIC 9(6).

              The network trace number of the transaction.

230   [94]    tstmh^data.bnk
```

```
Offset[Len]   Field Name and Description                       Data Type

              The following fields contain the institution and teller
              terminal identification data.

230   [4]     tstmh^data.bnk.regn^id.byte[0:3]
                                [REGN]                          PIC X(4).

              The region ID associated with the terminal.

234   [4]     tstmh^data.bnk.brch^id.byte[0:3]
                                [BRCH]                          PIC X(4).

              The branch ID associated with the terminal.

238   [25]    tstmh^data.bnk.term^loc.byte[0:24]              PIC X(25).

              The address or location description of the terminal
              where the transaction was initiated.

263   [13]    tstmh^data.bnk.term^city.byte[0:12]             PIC X(13).

              The city of the terminal where the transaction was
              initiated.

276   [3]     tstmh^data.bnk.term^st.byte[0:2]                 PIC X(3).

              The state, province, or political division of the
              terminal where the transaction was initiated.  In the
              United States, only the first two characters are used
              to identify the state where the terminal is located.
              Codes for other countries must be agreed upon by all
              sharing institutions.

279   [2]     tstmh^data.bnk.term^cntry.byte[0:1]              PIC X(2).

              The country of the terminal where the transaction was
              initiated.  Valid values are listed in the Codes
              for the Representation of Names of Countries, ISO
              3166-1988.

281   [11]    tstmh^data.bnk.acq^inst^id^num.byte[0:10]
                                [ID-NUM]                       PIC 9(11).

              The institution id number of the terminal owner
              if the terminal is directly connected to BASE24-
              teller.  Otherwise the id number of the acquiring
              entity introducing the transaction into BASE24.
              Data is right-justified and zero-filled on the left.

292   [11]    tstmh^data.bnk.rcv^inst^id^num.byte[0:10]
                                [ID-NUM]                       PIC 9(11).

              The institution id number of the account owner
              or card issuer.  Data is right-justified and zero-
              filled on the left.
```

```
Offset[Len]   Field Name and Description                    Data Type


303   [1]     tstmh^data.bnk.dda^cur^flg                    PIC X(1).

              A code specifying if the PBF for DDA accounts is
              current.  Valid values are:

              Y = Yes, the file is current.
              N = No, the file is not current.

304   [1]     tstmh^data.bnk.sav^cur^flg                    PIC X(1).

              A code specifying if the PBF for savings accounts is
              current.  Valid values are:

              Y = Yes, the file is current.
              N = No, the file is not current.

305   [1]     tstmh^data.bnk.ccd^cur^flg                    PIC X(1).

              A code specifying if the PBF for credit accounts is
              current.  Valid values are:

              Y = Yes, the file is current.
              N = No, the file is not current.

306   [1]     tstmh^data.bnk.spf^cur^flg                    PIC X(1).

              A code specifying if the SPF and PBF are current.  Valid
              values are:

              Y = Yes, the files are current.
              N = No, the files are not current.

307   [1]     tstmh^data.bnk.nbf^cur^flg                    PIC X(1).

              A code specifying if the NBF and PBF are current.
              Valid values are:

              Y = Yes, the files are current.
              N = No, the files are not current.

308   [1]     tstmh^data.bnk.whff^cur^flg                   PIC X(1).

              A code specifying if the WHFF and PBF are current.
              Valid values are:

              Y = Yes, the files are current.
              N = No, the file are not current.

309   [11]    tstmh^data.bnk.bnk^rtg^cde.byte[0:10]        PIC X(11).

              The bank routing code that may be entered by the teller
              in order to identify the institution that owns the
              account.  If entered, this routing code is used to
```

```
Offset[Len]   Field Name and Description                       Data Type

              locate the appropriate information from the Account
              Routing File (ARF).  The bank routing code is right-
              justified and zero-filled.

320   [1]     tstmh^data.bnk.bnk^relnshp                       PIC X(1).

              The banking relationship defined for the terminal
              owner and account owner.


321   [3]     tstmh^data.bnk.crncy^cde.byte[0:2]               PIC 9(3).

              The currency code used for the transaction.

324   [120]   tstmh^data.rqst

              The following fields contain the transaction-specific
              processing data.

324   [6]     tstmh^data.rqst.dev^tran^cde.byte[0:5]           PIC X(6).

              The device transaction code from the device native
              message.  The contents of this field varies for each
              device type and transaction code.

330   [8]     tstmh^data.rqst.cust^passbook^bal
                                                  TYPE BINARY 64 SIGNED.

              The customer’s passbook balance entered by the teller
              and used for passbook printing.

338   [1]     tstmh^data.rqst.auth^lvl                         PIC X(1).

              A code indicating the authorizor of the transaction
              from the perspective of the account owner.  Valid
              values are:

              0 = Transaction Not Supported
              1 = Online Authorization, Host Authorizes
              2 = Offline Authorization, BASE24 Authorizes
              3 = Reserved for future use
              4 = Log Only, BASE24-teller Authorization Process
                  logs transaction to the TTLF

              The Positive Balance File (PBF) on BASE24 is only
              updated for transactions with an AUTH-LVL value of 2
              (Offline Authorization).

339   [40]    tstmh^data.rqst.track2.byte[0:39]               PIC X(40).

              The Track 2 Data as received from the teller
              device.  The data may be read electronically or
              entered manually.  The data is in the following
              general format:
```

```
Offset[Len]   Field Name and Description                     Data Type

                  1) start sentinel (;)
                  2) PAN, left justified
                  3) a field separator (=)
                  4) expiration date (MMYY or YYMM)
                  5) discretionary data
                  6) end sentinel (?)

              If the data is manually entered, the device handler
              replaces the start sentinel with a value of M.

379   [19]    tstmh^data.rqst.from^acct
                                [ACCT]

              The from account number or the account number associated
              with the first account type.  If the from account number
              is not applicable or not available, this field is
              initialized to blanks.

379   [19]    tstmh^data.rqst.from^acct.acct^num.byte[0:18]
                                                            PIC X(19).

398   [19]    tstmh^data.rqst.to^acct
                                [ACCT]

              The to account number or the account number associated
              with the second account type.  If the to account number
              is not applicable or not available, this field is
              initialized to blanks.

398   [19]    tstmh^data.rqst.to^acct.acct^num.byte[0:18]   PIC X(19).

417   [6]     tstmh^data.rqst.tran

              The values in the following fields identify the type of
              transaction and from and to account types.  Refer to the
              DDL source file for the BASE24-teller Standard Internal
              Message Header (TSTMH) or the Teller Transaction File
              (TTF) for a complete list of the transaction codes and
              account types supported by BASE24-teller.

417   [2]     tstmh^data.rqst.tran.cde.byte[0:1]             PIC X(2).


419   [2]     tstmh^data.rqst.tran.from^acct^typ.byte[0:1]   PIC X(2).

              The from account type of the transaction.

421   [2]     tstmh^data.rqst.tran.to^acct^typ.byte[0:1]     PIC X(2).

              The to account type of the transaction.

423   [1]     tstmh^data.rqst.intl^ovrrd^lvl                 PIC X(1).
```

```
Offset[Len]   Field Name and Description                       Data Type


              A code indicating the initial override level used to
              process the transaction.  Valid values are:

              0 = No override level--The transaction was
                  processed without a particular override level.
              1 = Teller override level--The transaction was
                  processed with an override level of a teller.
              2 = Supervisor override levle--The transaction was
                  processed with an override level of a supervisor.
              3 = Manager override level--The transaction was
                  processed with an override level of a manager.

424   [1]     tstmh^data.rqst.max^term^ovrrd^lvl               PIC X(1).

              A code indicating the maximum override level that
              a particular terminal can handle.  For example,
              if the terminal has the capability of processing
              teller overrides (Level 1) and supervisor overrides
              (Level 2), this field should be set to a 2.
              Valid values are:

              1 = Teller level.  The terminal has the capability of
                  processing teller overrides.
              2 = Supervisor level.  This terminal has the capability
                  of processing teller and supervisor overrides.
              3 = Manager level.  This terminal has the capability
                  of processing teller, supervisor, and manager
                  overrides.

425   [1]     tstmh^data.rqst.paperless^tran                   PIC X(1).

              A code indicating whether the transaction is paperless
              or should be processed using a paper source document.
              The value in this field is not used by BASE24-teller,
              but is provided to assist the host software.  Valid
              values are:

              Y = Yes, the transaction is paperless and should be
                  posted to host files from the TTLF.
              N = No, the transaction is not paperless.  It should
                  be posted to host files from a paper source
                  document instead of the TTLF.

426   [1]     tstmh^data.rqst.compl^req                        PIC X(1).

              A code specifying whether BASE24 should send
              completion messages to the host for transactions.
              Valid values are:

              N = No, do not send completion messages to the host.
              A = Yes, send completion messages to the host for
                  approved transactions only.
              D = Yes, send completion messages to the host for
```

```
Offset[Len]   Field Name and Description                       Data Type

                  denied transactions only.
              B = Yes, send completion messages to the host for
                  approved and denied transactions.

427   [1]     tstmh^data.rqst.advc^resp^req                    PIC X(1).

              A code specifying whether a response is required for a
              teller-initiated advice or reversal advice message.
              Valid values are:

              0 = No response is required.
              1 = Yes, a response is required.

428   [1]     tstmh^data.rqst.crd^present                      PIC X(1).

              A code indicating whether the transaction was initiated
              by a card at the teller terminal.  Valid values are:

              Y = Yes, the transaction was initiated by a card.
              N = No, the transaction was not initiated by a card.

429   [3]     tstmh^data.rqst.entry^mde.byte[0:2]              PIC X(3).

              A code indicating the entry mode of the teller
              transaction.  It contains both how the card
              information was entered (positions 1 and 2) and the
              device PIN entry capability (position 3).  Valid
              values are:

                  Position 1-2
                  00    - Unspecified
                  01    - Manual
                  02    - Magnetic Stripe
                  03    - Bar Code
                  04    - OCR
                  05    - Integrated Circuit Card
                  06-60 - Reserved for ISO use
                  61-80 - Reserved for National use
                  81-99 - Reserved for Private use

                  Position 3
                  0     - No PIN entry capability
                  1     - PIN entry capability
                  2-5   - Reserved for ISO use
                  6-7   - Reserved for National use
                  8-9   - Reserved for Private use

432   [2]     tstmh^data.rqst.rsn^cde.byte[0:1]                PIC X(2).

              An informational field which can be used to distinquish
              between different types of the same transaction.  This
              field is frequently used for Miscellaneous Debit/Credit
              Transactions and Debit/Credit Memo Post Transactions to
              indicate the specific reason for the transaction.  The
```

```
Offset[Len]   Field Name and Description                   Data Type

              values associated with this field are user-defined and
              may vary for each BASE24-teller customer.

434   [1]     tstmh^data.rqst.mult^acct                    PIC 9(1).

              A code indicating whether the terminal is capable of
              handling multiple account data.  Valid values are:

              0 = Terminal does not support; default to primary
                  account
              1 = Terminal supports; return account selection
                  information

435   [8]     tstmh^data.rqst.ovrrd^tlr^id.byte[0:7]       PIC X(8).

              The identification of the teller, supervisor or manager
              that overrode the transaction.

443   [1]     tstmh^data.rqst.user^fld2                    PIC X(1).

              Reserved for future use.

444   [2]     tstmh^data.tim^ofst              TYPE BINARY 16 SIGNED.

              The time difference between the transaction-initiating
              terminal and the Tandem processor location.  It is the
              signed number of minutes to be added to the Tandem
              system time in order to obtain the terminal time.

446   [46]    tstmh^data.resp^hdr

              The following fields contain the transaction response
              data.

446   [1]     tstmh^data.resp^hdr.crd^action               PIC X(1).

              A code representing the return/retain card action that
              was set on the original response prior to the
              override.  This field is only used for card-initiated
              transactions.  Valid values are:

              0 = Return Card
              1 = Retain Card

447   [3]     tstmh^data.resp^hdr.resp^cde.byte[0:2]       PIC X(3).

              The response code associated with the transaction.
              Refer to the DDL source file for the BASE24-teller
              Standard Internal Message Header (TSTMH) or the
              Override Response File (ORF) for a complete
              list of the response codes supported by BASE24-teller.

450   [6]     tstmh^data.resp^hdr.post^dat
```

```
Offset[Len]   Field Name and Description                     Data Type


              The date (YYMMDD) the transaction is posted by the
              financial institution.

450   [2]     tstmh^data.resp^hdr.post^dat.yy.byte[0:1]      PIC X(2).

452   [2]     tstmh^data.resp^hdr.post^dat.mm.byte[0:1]      PIC X(2).

454   [2]     tstmh^data.resp^hdr.post^dat.dd.byte[0:1]      PIC X(2).

456   [6]     tstmh^data.resp^hdr.auth^id^resp.byte[0:5]     PIC X(6).

              The authorization id response.  The value is not
              generated by BASE24-teller but may optionally be
              generated by the authorizing host.

462   [1]     tstmh^data.resp^hdr.err^flg                    PIC X(1).

              A code used to provide additional information regarding
              the disposition of the transaction.  Valid values are:

              C - Card Verification Failed
              K - KMAC Synchronization Error (future)
              M - MAC Failure (future)
              S - Sanity Check
              T - Token Error

463   [1]     tstmh^data.resp^hdr.min^ovrrd^lvl              PIC X(1).

              A code indicating the minimum level of override
              necessary to authorize the transaction in situations
              when the transaction is normally declined.  This level
              is defined in the Override Response File (ORF).
              Valid values are:

              0 = No override required--The transaction is approved
                  and does not require an authorization override.
              1 = Teller authorizes override--The transaction can be
                  approved if entered by a teller.
              2 = Supervisor authorizes override--The transaction can
                  be approved if entered by a supervisor.
              3 = Manager authorizes override--The transaction can
                  be approved if entered by a manager.
              4 = Override restricted--The transaction is denied and
                  cannot be approved with an authorization override.

464   [1]     tstmh^data.resp^hdr.acct^ind                   PIC X(1).

              A code used when a fatal response code is associated
              with a specific account (i.e. From, To or Credit Line).
              This indicator is set by the Authorization
              process on selected card related response codes
              (i.e. F32, F33, F34, F3C), account related response
              codes (i.e. F4x where x indicates a value from 0 to 9)
```

```
Offset[Len]   Field Name and Description                       Data Type

              stop payment related response codes (i.e. F5x),
              warning/hold/float related response codes (i.e. F6x) and
              passbook related response codes (i.e. F7x).  The account
              indicator may also be set in conjunction with response
              codes F08, F0B, F0C, O4x, O5x, O6x, and O7x.  The
              account indicator will only be set in conjunction with
              an overridable response code when the response code
              can not be overridden.  Valid values are:

              F = From Account
              T = To Account
              C = Credit/Backup Line Account

465   [1]     tstmh^data.resp^hdr.crd^vrfy^flg                 PIC X(1).

              A code indicating the card verification action
              performed by the Authorization process.  Valid values
              are:

              C = Card verification performed, CVD invalid, condition
                  noted, transaction authorization continued.

              D = Card verification performed, CVD invalid,
                  transaction denied.  ERR-FLG contains a value of
                  C.

              N or 0 = Card verification not attempted or security
                       device error.

              Y = Card verification performed, CVD valid.

466   [16]    tstmh^data.resp^hdr.user^fld4.byte[0:15]        PIC X(16).

              Reserved for future use.

482   [10]    tstmh^data.resp^hdr.user^fld5.byte[0:9]         PIC X(10).

              Reserved for future use.

492   [8]     tstmh^data.entry^tim               TYPE BINARY 64 SIGNED.

              The time, expressed in Greenwich Mean Time, the
              transaction entered the BASE24 system for the first
              time.  A Device Handler loads this field when the
              TSTMH is initialized.

500   [8]     tstmh^data.exit^tim                TYPE BINARY 64 SIGNED.

              The time, expressed in Greenwich Mean Time, the
              transaction left the BASE24 system for authorization
              through the Host Interface process.

508   [8]     tstmh^data.re^entry^tim            TYPE BINARY 64 SIGNED.
```

```
Offset[Len]   Field Name and Description                      Data Type

              The time, expressed in Greenwich Mean Time, when
              the response to an original transaction re-enters
              the BASE24 system.

              END   [size   516]
```